

FutureSportler – AI / Computer Vision Engineer

Technical Assignment

Objective

Design and implement a robust computer vision pipeline capable of tracking all key entities in sports videos and projecting them onto a top-down 2D Hawk-Eye–style tactical map.

Tracked Entities

- Players
- Ball
- Referee / Umpire

Sport Selection

Candidates must choose **ONLY ONE** sport from the following options:

- **Volleyball**
- **Cricket**

Input Videos

Volleyball Videos:

https://drive.google.com/drive/folders/1nTK95Lpn_KR5gthvCEie2nQbrFwzaINm

Cricket Videos:

https://drive.google.com/drive/folders/1kFXWTAGk_rM_z_gFEf34yF4GsUqyFQYx

Core Requirements

- Accurate detection of players, ball, and referee/umpire
- Continuous multi-object tracking with stable IDs
- Team classification for players
- 2D Hawk-Eye–style projection of all entities
- Frame-by-frame synchronized tactical map visualization

Submission Requirements

- GitHub repository containing complete source code
- README explaining approach, assumptions, and limitations
- Output videos (annotated video + 2D tactical map)
- Output videos **MUST** be submitted via a Google Drive link only

Deadline

All submissions must be completed by **27 December**.

Reference Material

<https://www.youtube.com/watch?v=yGQb9KkvQ1Q>

https://nihal111.github.io/hawk_eye/

<https://github.com/roboflow/sports>